

Track 3: FinTech AI

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2. Problem statement

The prototype addresses a timing problem reported in small-business surveys: a firm can be owed money by customers while payroll, rent, and other bills still fall due. Those survey results are cited below. They describe other firms. They are not measurements of Harbor & Co. Every dollar figure later in this report comes from this project's generator, forecast, and dashboard.

In the 2024 Small Business Credit Survey, 51% of small employer firms said uneven cash flow, including collecting on receivables, was a financial challenge in the prior 12 months, and 56% said the same about paying operating expenses such as payroll, rent, and inventory. Rising costs of goods, services, and wages were more common, at 75%. The survey was fielded from September to November 2024 and received 7,653 responses from firms with 1 to 499 employees. The Federal Reserve Banks state that it is a convenience sample, not a random sample [1].

The 2023 payments module of the same survey, published in December 2024, says that managing cash flow is one of the top financial challenges small businesses face and that customer payments are the primary source of cash available to a firm. Of the employer firms in that module, 28% said the arrangement accounting for the highest share of revenue was payment after delivery, for example by invoice. The report says slow-paying customers are a common challenge for firms that collect after delivery, on a set schedule, in instalments, or through a third party. About 80% of firms in the module reported at least one payments-related challenge. The module is optional, and the report says its results should be read with the limits of a convenience sample [2].

An online survey commissioned by Intuit QuickBooks, conducted in December 2025 with 1,305 US owners of businesses with 0 to 250 employees, reported that 59% had at least some invoices overdue by 30 days or more. In the same survey, 39% said that one late payment in the past year made it hard to cover payroll or bills [3]. The Harbor & Co. demo uses that kind of timing gap as a synthetic example. It does not reproduce these survey percentages.

FinSight AI is a prototype for one synthetic business, Harbor & Co. It forecasts daily cash from scheduled bills and historical payment delays, compares that forecast with a trailing-average baseline, explains the item that takes the path through zero, and recalculates the same engine when the user changes an assumption.

3. Problem understanding

Who it is for

The target user is the owner of a small business who can see recent invoices and expenses but does not have a forward cash calendar. In this prototype the business is Harbor & Co., a synthetic studio. The books are generated by a python script and stored as transactions.csv, invoices.csv, and meta.json.

What the books contain

1. 632 transactions from 1 September 2024 through 2 March 2026.
2. 54 invoices. One is unpaid on the as-of date: BR-109, Brightline Studio, USD 20,000, issued 18 February 2026 and due 10 March 2026.
3. Recurring outflows placed on fixed calendar days: rent USD 4,800 on day 1 (Oak Street Landlord), payroll USD 22,000 on day 23, a loan payment of USD 1,800 on day 25 (North Bank), and marketing USD 2,400 on day 18 (Ad Network).
4. Other spending: utilities, software, weekday supplies, and a weekly contractor payment to Lane Design.
5. Three repeating customers: Maple Goods (USD 7,200), Harbor Civic (USD 9,400), and Brightline Studio (USD 19,000 on earlier invoices). Brightline's historical delays cycle through 8, 14, 20, 26, and 32 days after the due date.
6. A Seasonal Retainer of USD 32,000 was collected on 28 February 2026. It is already paid, so it raises recent cash and the trailing average, and it does not fund March payroll.
7. On 18 February 2026 the generator adds an extra marketing spend of USD 4,500, labelled as a campaign overrun, on top of the regular USD 2,400.

Opening cash in meta.json is about USD 50,226 before 1 September 2024. That figure is chosen so that cash on 2 March 2026 equals the generator's target of USD 16,000.

The pain the prototype is built to show

Over the 90 days ending 2 March 2026, invoices issued total USD 130,400 and expenses total USD 116,063. Collected client payments in that window total USD 138,800. Invoiced activity is higher than expenses, and cash on hand is USD 16,000. The schedule-aware path still crosses zero on 23 March 2026, 21 days after the as-of date, because payroll of USD 22,000 lands before Brightline's open invoice is expected to clear.

4. Proposed solution

FinSight AI loads the synthetic books, forecasts cash for 90 days, flags the first day the base path is below zero, explains the scheduled item responsible, and lets the user move one open invoice's collection date or cut one discretionary expense. The explanation is assembled from the forecast path. There is no chatbot and no language model. It has five sections: Position, Forecast, Why, What-if, and Evaluation.

5. Innovation and uniqueness

The difference from a trailing average is that the forecast uses the cash calendar the baseline ignores.

- The baseline extends the average daily net cash of the last 30 days and does not place payroll, rent, loan, marketing, or open invoices on dates.
- The schedule-aware forecast starts from cash on the as-of date, places rent, payroll, loan, and marketing on their recent calendar days at the median recent amount, and places each open invoice on the due date plus that customer's median historical delay.
- Other daily spending is projected with a ridge regression. The uncertainty band resamples historical payment delays and the residuals of that regression.

On this sample, that difference is measurable. In the 60-day holdout before 2 March 2026, the schedule-aware forecast's mean absolute error on daily cash is USD 2,819. The trailing-average baseline's error is USD 17,847. Cash first closed below zero in that holdout on 23 January 2026. The schedule-aware forecast flagged that date 15 days ahead. The baseline did not flag it. These results apply only to this synthetic protocol.

6. Solution architecture

The same forecast function is used by the chart, the what-if controls, and the evaluation. A scenario does not call a second model. It calls forecast again with a collection-date override or an expense cut.

- generate.py builds and loads the books.
- forecast.py computes the daily history, the schedule-aware path, the baseline, the percentile band, and the shortfall date.
- simulate.py passes a new collection date or a percentage cut into that function.
- explain.py reads the scheduled items on the path and names the item that first takes the balance below zero. It also compares the last 30 days of non-scheduled spending with earlier 30-day windows.

- evaluate.py reruns both paths from weekly origins inside the holdout, without using data from after each origin.
- Plotly draws the charts.

Technology used to run the prototype: Python, pandas, NumPy, scikit-learn (Ridge), Streamlit, and Plotly. Dependencies are listed in requirements.txt and installed in a project virtual environment, .venv. There is no database. The books are CSV and JSON files.

7. AI and technology used

Schedule-aware cash path

For each of rent, payroll, loan, and marketing, the engine takes the median day of the month in history, capped at day 28, and the median of the last six amounts. Open invoices are collected at the due date plus the customer's median delay when that customer has at least two paid invoices. If the expected date is already past, collection is placed on the next day. Brightline Studio has 17 paid invoices before the as-of date, a median delay of 20 days, and a standard deviation of about 8.8 days. The dashboard rounds that spread to 9 days. BR-109 is therefore expected after its 10 March due date, not on the due date. The warning appears because the spread is at least 7 days and the invoice is at least USD 5,000.

Variable spend

Client payments and the four scheduled categories are excluded from variable spend. A ridge regression with alpha 1.0 predicts the remaining daily outflow from the day of the week, the previous day's variable outflow, and the outflow seven days earlier. The prediction is not allowed to be negative.

Uncertainty band

The base dashboard draws 200 simulated paths with NumPy's default generator and seed 7. Each path resamples a historical delay for each open invoice and adds normal noise, using the residual standard deviation of the ridge fit, to variable spend. The chart shows the 10th and 90th percentiles. The "paths below zero within 30 days" figure is the share of those paths whose minimum over the next 30 days is negative. On the base case that share is 100 percent. The what-if control uses 80 simulations. Evaluation turns simulations off so the error score uses the base path only.

What the prototype does not use

There is no large language model, no chatbot, and no claim that the band is a statistical guarantee. The assumptions printed in the app are the five statements returned by the forecast: scheduled bills continue at recent median amounts, open invoices use the median delay unless a scenario

sets the date, other spending uses the ridge regression, the band is a range of simulated paths, and scenario results are calculations under those assumptions.

8. Prototype and model output

The screenshots below are from https://finsight-off.vercel.app/demo.
Dollar figures on the screen are rounded to the nearest dollar.

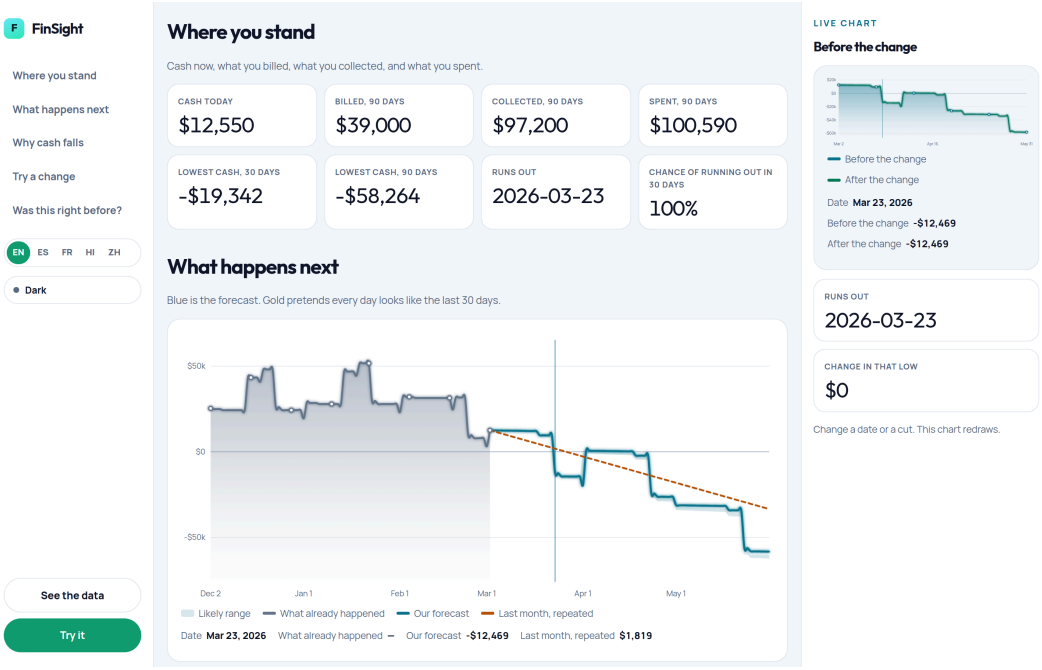


Figure 1. Position and the start of the forecast. Cash today is USD 16,000. The shortfall date is 23 March 2026.

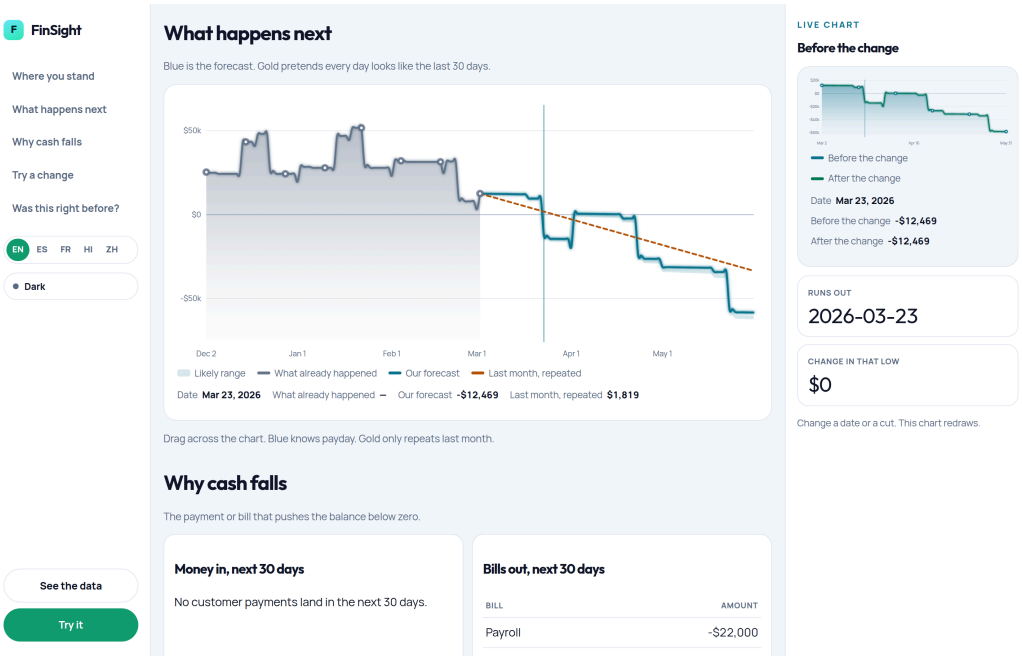


Figure 2. Forecast and the Why section. The dashed line is the trailing average. The dotted vertical line marks the shortfall.

The Why section lists one inflow in the next 30 days: Brightline Studio, USD 20,000. Scheduled outflows in that window are payroll USD 22,000, rent USD 4,800, marketing USD 2,400, and the loan USD 1,800. The engine reports that payroll of USD 22,000 on 23 March 2026 is the scheduled item that takes cash through zero, to about USD −12,469. The lowest base balance in 30 days is about USD −15,771. The lowest in 90 days is about USD −70,510.

The same section reports that marketing spending over the last 30 days is USD 6,900, against a prior 30-day average of USD 2,400 that did not vary. That USD 6,900 is the regular USD 2,400 plus the USD 4,500 overrun written into the February books. The check is a comparison with earlier windows, not a separate anomaly model. It fires when the recent total is above 1.5 times a prior average that had no spread, or more than two standard deviations above a prior average that did vary.

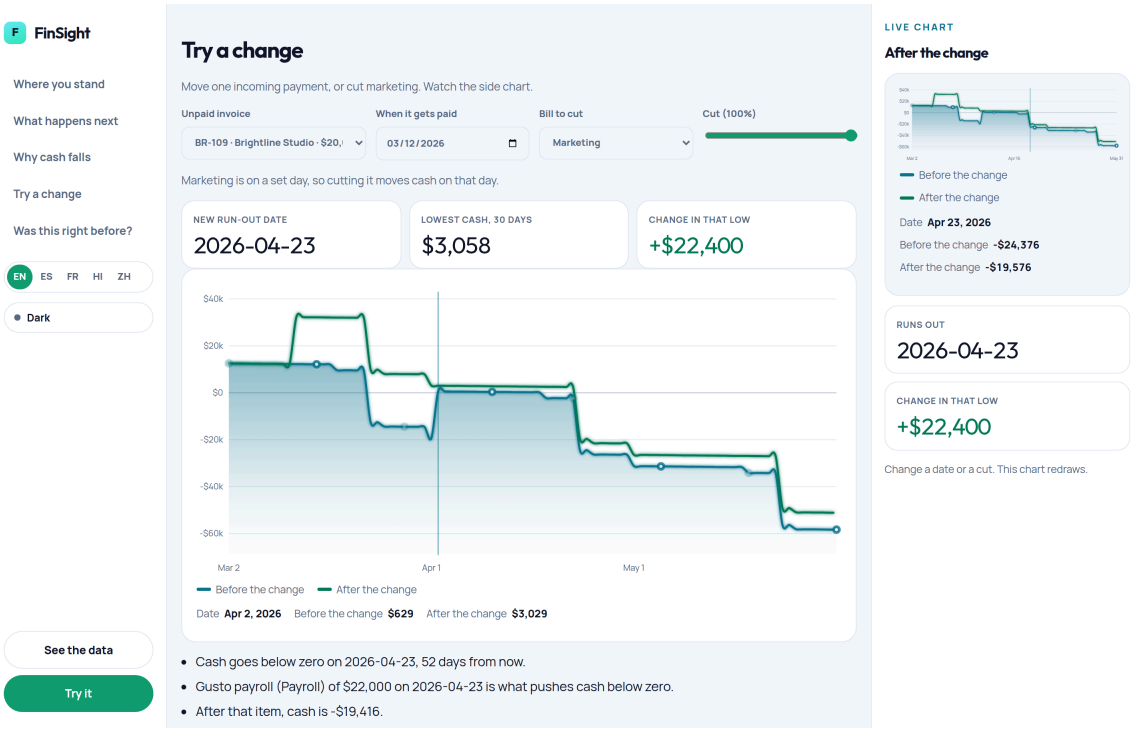


Figure 3. What-if with BR-109 collected on 12 March 2026 and marketing reduced by 100 percent.

With those two changes, the screen shows a scenario shortfall of 10 April 2026, a 30-day low of USD 1,562, and a change in the 30-day low of USD 17,334 versus the base path. The scenario text says the path drops below zero 39 days out. Running the same function with only the collection date moved to 12 March 2026, and no expense cut, moves the shortfall to 1 April 2026, raises the 30-day low by about USD 14,934, and leaves that 30-day low slightly negative, about USD −838. The extra USD 2,400 between the two 30-day-low changes is the March marketing bill that the 100 percent cut removes. These are recalculations of the cash engine, not evidence that a customer will pay on the chosen date.

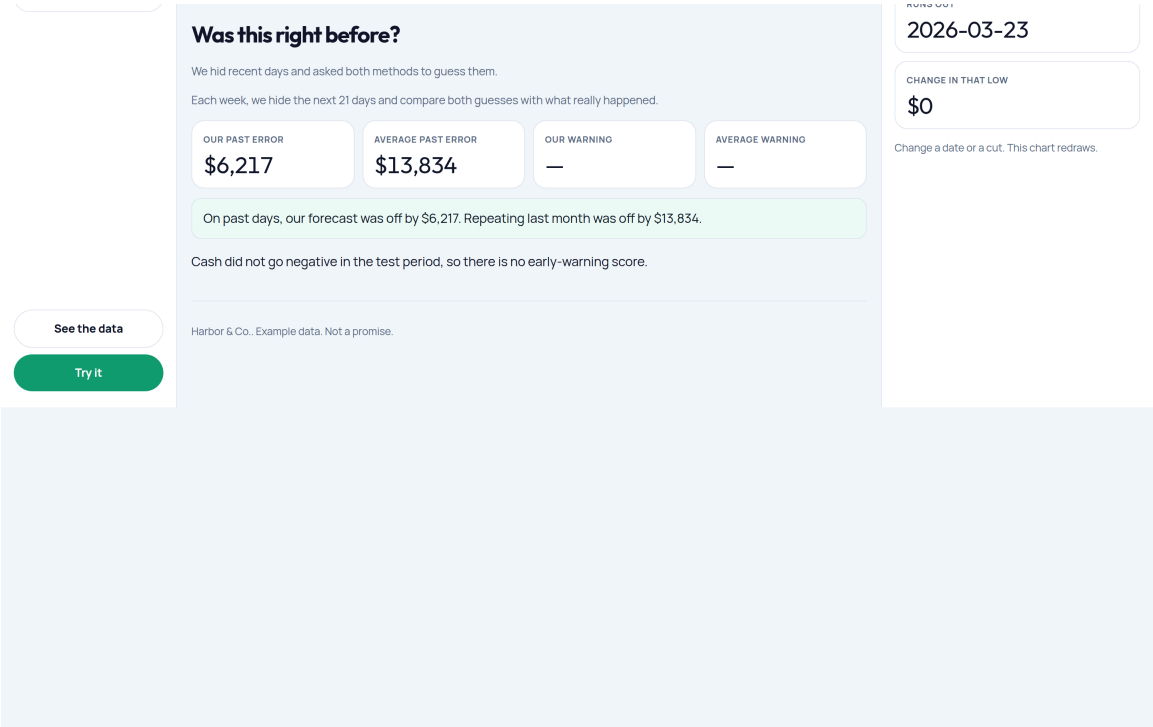


Figure 4. Holdout scores shown in the Evaluation section.

Evaluation uses weekly origins in the 60 days before 2 March 2026. Each origin forecasts the next 21 days, using only transactions and invoices known on that origin. Payments dated after the origin are treated as still open. There are 6 origins. Error is the mean absolute difference from the actual daily closing cash. Lead time is how many days before the holdout shortfall a method first projected a below-zero balance on or before that date.

Measure	Schedule-aware forecast	Trailing-average baseline
Mean absolute error	USD 2,819	USD 17,847
Holdout shortfall	23 January 2026	23 January 2026
Lead time	15 days	Not flagged
Origins × horizon	6 origins, 21 days	Same origins

Five unit tests in tests/test_cash.py passed when run with the project virtual environment. They check a hand-built shortfall date, the effect of moving a payment, and that evaluation returns both methods. They do not replace the holdout numbers above.

9. Business impact

On this synthetic business, the prototype shows a cash gap that a 90-day invoice-versus-expense comparison does not show: invoices of USD 130,400 exceed expenses of USD 116,063, and cash is still projected to cross zero on 23 March 2026. It also shows which single scheduled item causes that crossing, and how much the 30-day low moves if BR-109 is collected on 12 March instead of on the historical-delay date.

The report does not claim a cost saving, a revenue increase, or a result on real bank data. No live business has been scored. The only measured improvement is the lower holdout error and the earlier shortfall flag versus the trailing average, on these generated books.

10. Feasibility

The prototype runs locally. Setup is a virtual environment, pip install -r requirements.txt, and streamlit run app.py with PYTHONPATH set to src. The first load writes the books if meta.json is missing.

Implementation limits that are visible in the code:

- One synthetic company. Categories, amounts, and the 21-day story are written in the generator.
- Recurring bills are assumed to continue. There is no feed from a bank or accounting system.
- Customers with fewer than two paid invoices fall back to a 10-day delay and a 5-day spread.
- The ridge model is a small lag model for residual spending. It is not a general demand forecast.
- The percentile band depends on the random seed and the number of simulations.
- A 90-day path that keeps falling, as the chart does into May, means later months are still short under the same assumptions. The prototype does not add new sales to refill cash.

Scaling beyond the demo would require real transaction files, a definition of scheduled versus variable categories for each business, and an evaluation on held-out periods from those files. That work is not in this repository.

11. Future scope

These items are not implemented. They are the next uses of the same design.

- Replace the generated CSV files with a bank or invoice export in the same columns.
- Keep the baseline comparison whenever the data change, and show the result only if the schedule-aware error is still lower.

- Let the user add more than one collection-date change at a time. The engine already accepts a dictionary of overrides and a dictionary of expense cuts.
- State the shortfall probability next to the date when the simulated paths disagree with the base path. On this base case they agree: every simulated path is below zero within 30 days.

12. Conclusion

FinSight AI is a working cash-flow prototype for one synthetic business. On books that show USD 16,000 in the bank and more invoiced activity than expenses, it projects a shortfall on 23 March 2026, names payroll as the scheduled item that crosses zero, and shows the effect of collecting BR-109 earlier and cutting marketing. Against a trailing 30-day average, on a 60-day holdout with six weekly origins, its daily cash error is about USD 2,819 rather than about USD 17,847, and it flags the 23 January 2026 shortfall 15 days ahead. The baseline does not. Those statements are limited to this dataset and this protocol.

References

[1] Federal Reserve Banks. ″2025 Report on Employer Firms: Findings from the 2024 Small Business Credit Survey.” Small Business Credit Survey, 27 March 2025. https://doi.org/10.55350/sbcs-20250327. Also published at https://www.fedsmallbusiness.org/reports/survey/2025/2025-report-on-employer-firms.

[2] Federal Reserve Banks. ″2024 Report on Payments: Findings from the 2023 Small Business Credit Survey.” Small Business Credit Survey, December 2024. https://doi.org/10.55350/sbcs-20241205. PDF: https://www.fedsmallbusiness.org/-/media/project/clevelandfedtenant/fsbsite/reports/2024/2024-report-on-payments_sbcs.pdf.

[3] Intuit QuickBooks. ″2026 Small Business Late Payments Report.” https://quickbooks.intuit.com/r/small-business-data/small-business-late-payments-report-2026/. The page states that the late-payment findings cited here come from an online survey of 1,305 US business owners commissioned by Intuit QuickBooks in December 2025. Accessed 19 October 2026.

Presentation link:
https://docs.google.com/presentation/d/1EiWwH9nj9SIpUJljqd61KXQ5OEYh0K7u/edit?usp=sharing&ouid=109147068799916288237&rtpof=true&sd=true

Demo link: https://finsight-off.vercel.app/demo
GitHub repository: https://github.com/Nancy20052020/finsight_off
Deployed site link: https://finsight-off.vercel.app